SQL Code

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-- Project on SQL



-- Create Database
CREATE DATABASE OnlineBookstore;

-- Switch to the database
\c OnlineBookstore;


-- Create Tables
DROP TABLE IF EXISTS Books;

CREATE TABLE Books (
    Book_ID SERIAL PRIMARY KEY,
    Title VARCHAR(100),
    Author VARCHAR(100),
    Genre VARCHAR(50),
    Published_Year INT,
    Price NUMERIC(10, 2),
    Stock INT
);


DROP TABLE IF EXISTS customers;

CREATE TABLE Customers (
    Customer_ID SERIAL PRIMARY KEY,
    Name VARCHAR(100),
    Email VARCHAR(100),
    Phone VARCHAR(15),
    City VARCHAR(50),
    Country VARCHAR(150)
);



DROP TABLE IF EXISTS orders;

CREATE TABLE Orders (
    Order_ID SERIAL PRIMARY KEY,
    Customer_ID INT REFERENCES Customers(Customer_ID),
    Book_ID INT REFERENCES Books(Book_ID),
    Order_Date DATE,
    Quantity INT,
    Total_Amount NUMERIC(10, 2)
);

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SELECT * FROM Books;

SELECT * FROM Customers;

SELECT * FROM Orders;



-- Import Data into Books Table

COPY Books(Book_ID, Title, Author, Genre, Published_Year, Price, Stock)
FROM 'D:\code\SQL\SQL files\Books.csv'
WITH (FORMAT csv, HEADER true, DELIMITER ',');




-- Import Data into Customers Table
COPY Customers(Customer_ID, Name, Email, Phone, City, Country)
FROM 'D:\code\SQL\SQL files\Customers.csv'
CSV HEADER;




-- Import Data into Orders Table
COPY Orders(Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)
FROM 'D:\code\SQL\SQL files\Orders.csv'
WITH (FORMAT csv, HEADER true, DELIMITER ',');



SELECT * FROM Books;

SELECT * FROM Customers;

SELECT * FROM Orders;


-- Basic Questions

-- 1) Retrieve all books in the "Fiction" genre:

select * From Books
where genre = 'Fiction'


-- 2) Find books published after the year 1950:

select * From Books
where published_year > 1950


-- 3) List all customers from the Canada:
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select * From Customers
where country = 'Canada'


-- 4) Show orders placed in November 2023:

select * From Orders
where order_date between '2023-11-01' and '2023-11-30'


-- 5) Retrieve the total stock of books available:

select sum(stock) as total_stock
from Books


-- 6) Find the details of the most expensive book:

select * from Books
order by price Desc
limit 1


-- 7) Show all customers who ordered more than 1 quantity of a book:

select * from Orders
where quantity > 1


-- 8) Retrieve all orders where the total amount exceeds $20:

select * from Orders
where total_amount > 20


-- 9) List all genres available in the Books table:

select Distinct genre from Books


-- 10) Find the book with the lowest stock:

select * from Books
order by stock
limit 1


-- 11) Calculate the total revenue generated from all orders:

select sum(total_amount) as total_revenue
from Orders

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-- Advance Queries

-- 1) Retrieve the total number of books sold for each genre:

select b.Genre , sum(o.Quantity) as total_book_sold
from Orders o
join Books b
ON o.book_id = b.book_id
group by b.Genre ;

-- 2) Find the average price of books in the "Fantasy" genre:

select avg(price) as avg_price
from Books
where genre = 'Fantasy'

-- 3) List customers who have placed at least 2 orders:

select customer_id , count(order_id) as order_count
from Orders
Group by customer_id
Having count(order_id) >= 2

-- 4) Find the most frequently ordered book:

select book_id , count(order_id) as order_count
from Orders
group by Book_id
order by order_count desc
limit 1



-- to get the book name

select o.book_id , b.Title , count(o.order_id) as order_count
from Orders o
join Books b
on o.book_id = b.book_id
group by o.Book_id , b.Title
order by order_count desc
limit 1


-- 5) Show the top 3 most expensive books of 'Fantasy' Genre :
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select * from Books
where genre = 'Fantasy'
order by price desc
limit 3


-- 6) Retrieve the total quantity of books sold by each author:

select b.Author , sum(o.Quantity) as Total_book_sold
from Orders o
join Books b
on o.book_id = b.book_id
group by b.Author


-- 7) List the cities where customers who spent over $300 are located:

select distinct c.city , o.total_amount
from Orders o
join Customers c
On o.customer_id = c.customer_id
where o.total_amount >= 300


-- 8) Find the customer who spent the most on orders:

select distinct c.name , c.customer_id , sum(o.total_amount) as total_spent
from Orders o
join Customers c
On o.customer_id = c.customer_id
group by c.customer_id , c.name
order by total_spent desc
limit 20



--9) Calculate the stock remaining after fulfilling all orders:

select b.book_id , b.title , b.stock , coalesce(sum(o.quantity) , 0) as order_quantity
,
        b.stock - coalesce(sum(o.quantity) , 0 ) AS Remaining_quantity
from Books b
left join Orders o
on o.book_id = b.book_id
group by b.book_id
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266
267
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